SUPPLEMENTARY MATERIAL I

Appendix A

Full IBM Quantum Experimental Metadata and Protocols

A.1 Motivation and Central Prediction

This appendix documents the complete IBM Quantum experimental programme used to test the distinguishability framework. The objective of these experiments was not to test standard quantum mechanics, whose predictions remain fully consistent with all observed outcomes, but rather to investigate whether a higher-level geometric organization exists across physically distinct quantum-control protocols.

The central prediction of the distinguishability framework is that the measured excitation probability P_e is determined primarily by the distinguishability between the initial and final quantum states rather than by the detailed trajectory used to connect them. In this description, endpoint distinguishability acts as a universal organizing variable capable of unifying outcomes that would otherwise appear protocol-dependent.

More specifically, the framework predicts that quantum-control protocols possessing different pulse sequences, different Bloch-sphere trajectories, different accumulated path lengths, and even different hardware implementations should exhibit common behaviour when represented as functions of endpoint distinguishability d_{FS}.

The principal hypothesis tested throughout this appendix is therefore

$$P_e = f(d_{FS}),$$

where d_{FS} denotes the Fubini–Study distinguishability between the initial and final quantum states.

If correct, this prediction implies that endpoint distinguishability provides a more fundamental description of quantum-control outcomes than conventional laboratory control parameters such as pulse angle, pulse duration, accumulated trajectory length, or circuit history.

To test this hypothesis, a hierarchy of increasingly demanding experiments was performed, beginning with single-qubit protocols and extending to large-scale hardware-wide experiments involving up to 156 physical qubits distributed across multiple independent IBM Quantum processors.

A.2 Scope of the Experimental Programme

The experimental programme consisted of several complementary validation campaigns designed to probe different aspects of the distinguishability hypothesis.

The first group of experiments examined single-qubit protocols with substantially different control profiles. These tests investigated whether excitation probabilities collapse when represented as functions of endpoint distinguishability rather than conventional laboratory control parameters.

The second group examined Bloch-sphere trajectories possessing identical endpoint distinguishabilities but different accumulated path lengths ℓ_{FS}. These experiments tested whether the final outcome depends primarily on the endpoint geometry or on the detailed trajectory connecting the initial and final states.

The third group investigated history-dependent protocols containing closed loops in state space. Such experiments were designed to determine whether previously traversed paths contribute independently to the final excitation probability.

The fourth group extended the analysis to entangling two-qubit operations in order to determine whether the distinguishability description remains valid in the presence of quantum correlations.

The final validation campaigns involved large-scale hardware-wide experiments performed simultaneously on up to 127 and 156 physical qubits. These tests examined scalability, cross-backend transferability, and the possibility of hardware-independent quantum-control descriptions.

Together, these experiments provide a systematic investigation of distinguishability-based organization across more than two orders of magnitude in system size.

A.3 Experimental Platforms

All experiments were performed using IBM Quantum superconducting quantum processors belonging to the Heron generation.

The primary processors employed in this work were:

• IBM Kingston

• IBM Fez

• IBM Marrakesh

Each processor provided access to 156 physical superconducting qubits and was operated through the IBM Quantum cloud infrastructure.

The use of multiple independent processors was essential for testing whether the observed collapse behaviour represents a genuine geometric property of quantum evolution rather than a processor-specific artefact.

No backend-specific fitting parameters were introduced during the analysis. Whenever possible, protocols developed on one processor were executed directly on the others without additional optimisation. This allowed a direct test of hardware-independent transferability.

The experimental programme included:

• single-qubit protocols;

• Bloch-sphere trajectory tests;

• history-loop experiments;

• two-qubit entangling protocols;

• 127-qubit hardware-wide benchmarks;

• 156-qubit falsification tests;

• cross-backend transfer experiments;

• geodesic optimal-control experiments.

The complete archive of backend identifiers, execution records, and job metadata is provided below.

A.4 Software Environment

All experiments were executed using the IBM Quantum cloud infrastructure through Qiskit Runtime.

The archived software environment was:

Python version: 3.12.13

Qiskit version: 2.4.1

Qiskit IBM Runtime version: 0.47.0

Platform: Linux x86_64

The complete software environment was preserved together with the experimental datasets to ensure reproducibility of all reported results. The archived metadata include backend characteristics, runtime versions, execution timestamps, and environment information required for independent verification of the experimental procedures.

A.5 IBM Quantum Job Archive

Table A.1 summarises the principal IBM Quantum jobs used in the distinguishability study.

Table A.1. Principal IBM Quantum jobs used in the experimental validation programme.

Experiment	**Backend**	**Job ID**	**Qubits**
One-qubit protocol family	ibm_kingston	d8gq7nhe8nrc73bg8i00	1
Bloch-sphere path test	ibm_kingston	d8gpgtv8cos73f3i28g	5
History-loop test	ibm_kingston	d8g7nhe8nrc73bgc960	5
127-qubit distinguishability benchmark	ibm_kingston	d8hcie5v8cos73f48ma0	127
156-qubit cross-backend transfer test	ibm_kingston	d8hd21s2upec739ktc30	156
156-qubit geodesic optimal-control test	ibm_marrakesh	d8hdkphe8nrc73bh2u40	156

Additional execution records, backend metadata, calibration information, and auxiliary datasets were archived throughout the study and form part of the complete reproducibility package associated with this work.

The following sections analyse the experimental results obtained from these datasets and compare the predictive power of conventional control variables with the endpoint distinguishability description proposed in the main text.

A.6 One-Qubit Distinguishability Validation

The first validation stage examined whether endpoint distinguishability provides a better description of excitation probabilities than conventional laboratory control parameters.

A family of single-qubit protocols was executed on IBM Quantum hardware using substantially different pulse amplitudes, durations, and control profiles. For each protocol, the excitation probability P_e was measured and compared with several candidate organizing variables, including the laboratory control angle θ, the accumulated Fubini–Study path length ℓ_{FS}, and the endpoint distinguishability d_{FS}.

The distinguishability framework predicts that protocols terminating at the same endpoint distinguishability should produce similar excitation probabilities even when their control trajectories differ significantly. Conventional descriptions based on laboratory parameters do not generally predict such behaviour.

The experimental results demonstrated a clear collapse of the measured excitation probabilities when represented as functions of d_{FS}. In contrast, significantly larger scatter was observed when the same data were represented as functions of the laboratory control angle.

The one-qubit experiments therefore provided the first empirical indication that endpoint distinguishability acts as a more universal organizing variable than conventional control parameters.

A.7 Bloch-Sphere Path Test

The second validation stage investigated whether the measured excitation probability depends on the detailed trajectory followed by the quantum state on the Bloch sphere.

Protocols were constructed with nearly identical endpoint distinguishabilities d_{FS} but substantially different accumulated path lengths ℓ_{FS}. The objective was to determine whether the final outcome depends primarily on the endpoint geometry or on the specific path connecting the initial and final states.

If accumulated path length were the dominant variable, protocols possessing larger values of ℓ_{FS} would be expected to produce systematically different excitation probabilities. The distinguishability framework instead predicts that the dominant dependence should remain controlled by the endpoint distinguishability.

The experimental results strongly favoured the latter interpretation. Protocols possessing nearly identical values of d_{FS} produced nearly identical excitation probabilities despite significant differences in accumulated path length.

These observations indicate that the final physical outcome is largely determined by the endpoint geometry of the quantum evolution, whereas the accumulated trajectory length primarily influences implementation cost rather than the measured excitation probability itself.

A.8 History-Loop Test

A more stringent test of the distinguishability hypothesis was provided by history-dependent protocols containing closed loops in state space.

In these experiments, the quantum state was intentionally driven through additional loops before reaching the same final state. The purpose was to determine whether previously traversed trajectories contribute independently to the final excitation probability.

Conventional intuition might suggest that increasingly complicated histories could leave measurable traces in the final outcome. The distinguishability framework predicts otherwise. Because the endpoint distinguishability depends only on the initial and final states, protocols possessing identical endpoints should exhibit identical excitation probabilities regardless of intermediate loops.

The experimental data supported this prediction. Protocols containing substantially different history structures produced nearly identical excitation probabilities whenever the endpoint distinguishability remained unchanged.

The history-loop experiments therefore provide direct evidence that the measured excitation probability is primarily controlled by endpoint geometry rather than by the detailed history of the evolution.

A.9 Two-Qubit Entangling Validation

The distinguishability hypothesis was next tested in the presence of entanglement.

A set of two-qubit protocols involving entangling operations was executed on IBM Quantum hardware. These experiments were designed to determine whether the endpoint distinguishability description remains valid beyond the single-qubit regime.

For each protocol, the measured excitation probability was compared with several candidate organizing variables. As in the one-qubit experiments, the endpoint distinguishability consistently provided the strongest collapse of the data.

The persistence of the collapse behaviour in the presence of entanglement indicates that the distinguishability description is not restricted to simple single-qubit trajectories but extends naturally to multi-qubit quantum evolutions.

This result is particularly important because entanglement introduces additional geometric and dynamical structure not present in single-qubit systems. The continued success of the distinguishability description therefore provides evidence that the organizing principle remains applicable in more complex quantum systems.

A.10 Collapse-Metric Comparison

A direct quantitative comparison was performed between several candidate organizing variables.

The mean absolute collapse errors obtained from the experimental datasets are summarized in Table A.2.

Table A.2. Comparison of collapse quality for different organizing variables.

Organizing variable	**Mean absolute error**
Laboratory angle θ	0.1098
Path length ℓ_{FS}	0.0606
Endpoint distinguishability d_{FS}	0.0141

The results demonstrate that endpoint distinguishability provides the smallest collapse error among all tested variables.

Relative to the laboratory control angle, the distinguishability description reduces the collapse error by approximately a factor of eight. Relative to accumulated path length, the improvement is approximately a factor of four.

The superiority of d_{FS}becomes even more evident when extended collapse metrics are considered.

Table A.3. Maximum collapse errors for several candidate variables.

Variable	**Maximum error**
θ	0.304
ℓ_{FS}	0.496
d_{FS}	0.0566
T_G(path-based)	0.496
T_G(endpoint-based)	0.0566

These results constitute one of the strongest quantitative findings of the experimental programme. Across all tested protocols, endpoint distinguishability consistently provides the most accurate and hardware-independent organization of the measured excitation probabilities.

A.11 Endpoint Geometric Time Analysis

The distinguishability framework predicts that endpoint geometric time provides a natural extension of the endpoint distinguishability description. While the quantity d_{FS}measures the geometric separation between the initial and final states, the geometric time T_G characterizes the accumulated distinguishability evolution associated with the protocol.

To investigate this prediction, additional IBM Quantum experiments were analyzed using both path-based and endpoint-based definitions of geometric time. The objective was to determine whether the collapse observed for d_{FS} extends naturally to T_G.

The analysis demonstrated a clear distinction between the two constructions. Path-based geometric time produced collapse errors comparable to those obtained from accumulated path length ℓ_{FS}. In contrast, endpoint-based geometric time exhibited collapse quality nearly identical to that obtained using endpoint distinguishability itself.

The corresponding maximum collapse errors are summarized in Table A.3. The endpoint-based quantity yielded a maximum error of approximately 0.0566, whereas the path-based construction produced substantially larger deviations.

These observations suggest that the experimentally relevant quantity is associated primarily with endpoint geometry rather than with the detailed trajectory followed during the evolution. The result is fully consistent with the endpoint distinguishability principle developed throughout the main text.

A.12 127-Qubit Hardware-Wide Benchmark

The first large-scale validation of the distinguishability framework was performed using 127 physical qubits simultaneously on IBM Heron processors. The purpose of this experiment was to determine whether the collapse behaviour observed in few-qubit systems remains valid at the scale of an entire quantum processor.

Three protocol families were executed across multiple control parameters and on multiple independent processors. For each protocol, the endpoint distinguishability d_{FS} was calculated and compared with the measured mean excitation probability averaged over all participating qubits.

The resulting datasets obtained from independent processors collapsed onto a common dependence when represented as functions of endpoint distinguishability. No backend-specific fitting parameters were introduced and no processor-specific recalibration was required for the comparison.

The observed overlap demonstrates that endpoint distinguishability acts as a hardware-transferable organizing variable rather than a processor-specific quantity. The collapse quality remained comparable to that observed in the single-qubit and two-qubit experiments despite the increase of system size by more than two orders of magnitude.

Three independent IBM Quantum processors participated in the benchmark:

• IBM Kingston

• IBM Fez

• IBM Marrakesh

The corresponding hardware-wide benchmark jobs were:

Table A.4. IBM Quantum jobs used for the 127-qubit benchmark.

Backend	Job ID
ibm_kingston	d8hcie5v8cos73f48ma0
ibm_fez	d8hcii9e8nrc73bh1m1g
ibm_marrakesh	d8hciq5v8cos73f48mq0

The significance of this result extends beyond the specific distinguishability framework. It demonstrates that geometric descriptions formulated in terms of state-space distances remain meaningful even at the scale of more than one hundred simultaneously operating physical qubits.

A.13 156-Qubit Falsification Benchmark

A dedicated falsification benchmark was designed to test whether conventional laboratory control parameters can serve as universal organizing variables.

The benchmark employed the full 156-qubit capacity of IBM Heron processors and was specifically constructed to distinguish between two competing hypotheses:

$$P_e = f(\theta)$$

and

$$P_e = f(d_{FS}).$$

Three protocols were constructed with identical laboratory control angle $\theta = \pi$ but substantially different endpoint distinguishabilities.

Table A.5. Falsification benchmark with identical laboratory angle and different endpoint distinguishabilities.

Protocol	d_{FS}	**Measured** P_1
A_{direct}	1.570796	0.976519
B_{detour}	0.523599	0.267102
C_{loop}	0.785398	0.508395

Note that the endpoint distinguishability is determined by the final quantum state and is therefore not constrained to equal the laboratory control angle.

The results provide a direct falsification of the hypothesis that excitation probability is determined solely by the laboratory control angle.

Despite identical values of $\theta = \pi$, the measured excitation probabilities span nearly the full available probability range, from approximately 0.27 to 0.98.

The ratio

$$\frac{P_1(A)}{P_1(B)} \approx 3.66$$

demonstrates that laboratory control angle alone cannot provide a universal description of the observed outcomes.

Conversely, when represented as functions of endpoint distinguishability, the data collapse onto a common dependence consistent with the predictions of the distinguishability framework.

This experiment constitutes one of the strongest direct tests reported in this work because it was designed specifically to discriminate between competing organizing principles rather than merely demonstrate correlation.

Furthermore, the result was reproduced on multiple IBM Quantum processors, indicating that the observed behaviour is not associated with a particular hardware implementation.

Taken together, the falsification benchmark provides direct experimental evidence that endpoint distinguishability captures information not contained in conventional laboratory control parameters and therefore represents a more fundamental descriptor of quantum-control outcomes.

A.14 Cross-Backend Transferability

A central requirement of any physically meaningful organizing variable is transferability across independent hardware implementations. If the observed collapse behaviour were caused primarily by processor-specific calibration parameters, coupling maps, gate fidelities, or hardware imperfections, then protocols developed on one quantum processor would not be expected to predict outcomes on another processor without substantial recalibration.

To test this possibility, identical protocol families were executed on multiple IBM Heron processors without backend-specific optimisation. The objective was to determine whether endpoint distinguishability provides a hardware-independent description of the measured excitation probabilities.

The Kingston processor was used as the reference platform. Predictions obtained from the Kingston datasets were then compared directly with measurements performed on IBM Fez and IBM Marrakesh.

The cross-backend RMSE was computed pointwise in endpoint distinguishability. Kingston was used as the reference backend. For each matched value $d_i$of endpoint Fubini–Study distinguishability, the excitation probability measured on the target backend was compared with the Kingston reference value at the same d_i. The reported error is

$$\mathrm{RMSE}_{K\to B} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left[P_e^{(B)}(d_i) - P_e^{(K)}(d_i)\right]^2}.$$

where Kdenotes the Kingston reference backend, Bdenotes either Fez or Marrakesh, $d_i$are the matched endpoint distinguishability values, and $P_e^{(B)}(d_i)$ is the measured excitation probability on backend B. If the $d_i$grids were not identical, $P_e^{(K)}(d_i)$ was obtained by interpolation of the Kingston reference curve at the same d_i. When the sampled d_{FS}values were not identical between backends, the Kingston reference curve was linearly interpolated before computing the difference.

The resulting root-mean-square prediction errors were:

$$\mathrm{RMSE}(\mathrm{Kingston} \rightarrow \mathrm{Fez}) = 0.00465$$
$$\mathrm{RMSE}(\mathrm{Kingston} \rightarrow \mathrm{Marrakesh}) = 0.00671$$

Table A.6. Cross-backend transferability results.

Transfer	**RMSE**
Kingston → Fez	0.00498
Kingston → Marrakesh	0.00921

Both prediction errors remain below 1% over the full experimental range.

These results demonstrate that endpoint distinguishability provides a transferable description of quantum-control outcomes that does not require backend-specific fitting parameters. The same geometric quantity predicts experimental measurements on independent quantum processors with sub-percent accuracy.

The transferability test is particularly important because it separates geometric structure from hardware implementation. If the collapse depended strongly on processor-specific characteristics, the transfer errors would be expected to be substantially larger. The observed agreement therefore supports the interpretation that endpoint distinguishability captures a genuine geometric property of the quantum evolution rather than a calibration-dependent artefact.

A.15 Geodesic Optimal-Control Test

The final experimental campaign investigated whether the Fubini–Study geodesic provides the most efficient implementation among protocols connecting the same initial and final quantum states.

The distinguishability framework separates two conceptually different quantities:

• endpoint distinguishability d_{FS}, which determines the physical outcome of the protocol;

• accumulated path length ℓ_{FS}, which characterizes the geometric cost of the implementation.

To test this prediction, four protocol families were constructed. All protocols connected the same initial and final states and therefore possessed identical endpoint distinguishability. However, the accumulated Fubini–Study path lengths differed substantially.

Table A.7. Geodesic optimal-control benchmark.

Protocol class	ℓ_{FS}	**Implementation error**
Geodesic (shortest path)	1.57	0.0345
Short closed loop	3.14	0.0402
Medium closed loop	3.14	0.0409
Long closed loop	10.99	0.0551

The shortest geodesic protocol consistently produced the smallest implementation error.

As the accumulated path length increased, the implementation error increased systematically despite the fact that all protocols possessed the same endpoint distinguishability and therefore produced the same physical outcome.

This observation provides an important practical consequence of the distinguishability framework. While endpoint distinguishability determines the measured excitation probability, the accumulated path length governs the physical cost of implementing the protocol on real quantum hardware.

The results therefore suggest the following operational interpretation:

(i) endpoint distinguishability determines the physical outcome;

(ii) accumulated path length determines implementation cost;

(iii) the Fubini–Study geodesic provides the minimum-cost realization among protocols connecting the same initial and final states.

The geodesic optimal-control experiment thus establishes a direct connection between distinguishability geometry and practical quantum-control optimization.

A.16 Reproducibility Statement

All experiments reported in this work were executed through the IBM Quantum cloud infrastructure and archived together with the corresponding metadata.

The archived materials include:

• backend identifiers;

• IBM Quantum job identifiers;

• software environment metadata;

• calibration information;

• one-qubit datasets;

• Bloch-sphere trajectory datasets;

• history-loop datasets;

• two-qubit entangling datasets;

• 127-qubit benchmark datasets;

• 156-qubit falsification datasets;

• cross-backend transferability datasets;

• geodesic optimal-control datasets.

The software environment used for execution was archived together with the datasets and included Python 3.12.13, Qiskit 2.4.1, and Qiskit IBM Runtime 0.47.0.

The complete experimental archive contains the information necessary to reproduce all figures, tables, and quantitative results reported in the IBM Quantum section of this work.

Taken together, the experimental programme spans single-qubit, multi-qubit, and hardware-wide quantum-control protocols involving up to 156 simultaneously operating physical qubits on multiple independent IBM Quantum processors. The results consistently support the conclusion that endpoint distinguishability d_{FS}provides the most accurate and transferable geometric description of the measured excitation probabilities across physically distinct quantum-control protocols.

Appendix B

Derivation of the Gravitational Ramsey Dephasing Formula from a Single Principle

Atomic clocks work by counting the oscillations of an atom between two energy levels. If two identical clocks are placed at different heights — one on the floor, one on a shelf one centimetre higher - general relativity predicts that the higher clock ticks slightly faster, because gravity is weaker there. This is the gravitational redshift. Quantum mechanics then tells us that this frequency difference accumulates as a phase difference in the Ramsey interference pattern. In the standard textbook treatment, these two facts — the redshift and the phase — come from two completely separate theories and are combined by hand. What we show here is that both follow from a single geometric statement: the two clocks have accumulated different amounts of distinguishability. The formula is the same; the foundation is unified.

The Ramsey interference contrast of an atomic clock in a gravitational field is a well-known quantity, understood through the combination of gravitational redshift (from GR) and quantum phase evolution (from QM). The standard derivation requires two separate postulates: the Einstein redshift formula $\Delta\nu/\nu = gh/c^2$ imported from GR, and the Ramsey phase relation $\varphi = 2\pi\nu\cdot t$ imported from QM. These are then combined by hand. The present framework derives the same result from a single primitive structure - the equality of accumulated geometric times in the QM and GR realizations - without invoking either theory as a postulate.

Standard derivation (two postulates). Step 1- from GR: two clocks at heights h_1 and $h_2 = h_1 + h$ accumulate a frequency difference $\Delta\nu=\nu\cdot gh/c^2$ [65]. Step 2 - from QM: each clock acquires a Ramsey phase $\varphi = 2\pi\nu\cdot t$. The phase difference between the two clocks is $\Delta\varphi=2\pi\Delta\nu T=2\pi\nu(gh/c^2)T$. Step 3 - by hand: substitute the GR result into the QM formula. The two theories are glued externally. The formula $\Delta\varphi = 2\pi\nu(gh/c^2)T$ is therefore a product of two independent postulates, not a derivation from a unified structure.

B.1 Derivation from distinguishability geometry (one principle).

The single postulate is: physical evolution is characterized by the accumulation of distinguishable states, and geometric time $T_G = \int \mathcal{A}\, dt$ is the same quantity in all physical realizations. Two atoms at different heights h_1 and $h_2= h_1+h$ traverse different paths through the distinguishability manifold $\mathscr{F}$. We compute the accumulated geometric time for each realization and set them equal at the measurement event.

GR realization. The accessibility scale is $\Lambda_{GR} = c/\hbar$. The geodesic deviation speed between two worldlines separated by height h in Earth's gravitational field is $v_G^{GR} = gh/c$ (non-relativistic limit of the Jacobi equation). The accumulated GR geometric time over observation time T is:

$$\Delta T_G^{GR} = \int_0^T \frac{v_G^{GR}}{c}\, dt = \frac{gh}{c^2} \cdot T \tag{B1}$$

where: $v_G^{GR} = gh/c$ — geodesic deviation speed between two worldlines at height difference h in uniform gravitational field g; derived from the Jacobi equation $D^2\xi/d\tau^2 = -R^{\mu}_{\nu\rho\sigma}u^{\nu}\xi^{\rho}u^{\sigma}$ in the weak-field, slow-motion limit $\Lambda_{GR} = c/\hbar$ - GR accessibility scale, ΔT_G^{GR}— accumulated GR geometric time: the count of distinguishable geodesic-deviation states accumulated over T.

Here $D/D\tau$ denotes the covariant derivative taken along the reference worldline, τ is the proper time, $u^{\mu} = dx^{\mu}/d\tau$ is the four-velocity, and ξ^{μ} is the infinitesimal separation vector between neighbouring trajectories. The symbol $D/D\tau$ is used to distinguish covariant differentiation from ordinary differentiation $d/d\tau$. In curved spacetime, the evolution of vectors must be compared through parallel transport, which naturally leads to the covariant derivative.

QM realization. The accessibility scale is $\Lambda_{QM} = \Delta_n/\hbar$, where $\Delta_n = h\nu$ is the clock-transition energy gap. The Fubini–Study speed for a two-level atom accumulating Ramsey phase at rate $2\pi\nu$ is $v_{FS} = 2\pi\nu$ (one full Bloch-sphere revolution per period $1/\nu$ corresponds to arc length 2π in $\mathbb{P}(\mathscr{H})$). The accumulated QM geometric time is:

$$\Delta T_G^{QM} = \int_0^T \frac{v_{FS}}{\Delta_n/\hbar}\, dt = \frac{2\pi\nu}{2\pi\nu} \cdot T = T \tag{B2}$$

where: $v_{FS} = 2\pi\nu$ — Fubini–Study speed for resonant Ramsey evolution at clock frequency ν; equal to the angular velocity on the Bloch sphere, $\Delta_n = h\nu$ — clock-transition energy gap; the QM realization of the accessibility scale $\Delta T_G^{QM} = T$ — accumulated QM geometric time equals coordinate time for resonant evolution (one distinguishable state per unit time at ν)

Equality of geometric times. The phase difference measured in the Ramsey experiment is the difference in accumulated geometric times between the two arms of the interferometer, expressed in units of the QM clock period:

$$\varDelta\varphi = 2\pi \cdot \nu \cdot (\varDelta T_G^{GR} - \varDelta T_G^{QM}|_{baseline}) = 2\pi\nu \cdot \frac{gh}{c^2} \cdot T \tag{B3}$$

where: $\varDelta\varphi$ — Ramsey phase difference between the two arms; the observable measured by the interferometer , $2\pi\nu\ \varDelta T_G^{GR}$- contribution from gravitational distinguishability accumulation; converts GR geometric time to phase via clock frequency. The result $\Delta\varphi = 2\pi\nu(gh/c^2)T$ is identical to the standard result, obtained without invoking GR or QM as separate postulates

The standard derivation uses GR to get $\Delta\nu/\nu = gh/c^2$ and QM to get $\Delta\varphi = 2\pi \cdot \varDelta\nu \cdot T$, then combines them externally. The two theories are separate inputs. The present derivation uses a single object — the accumulated geometric time $T_G = \int \mathcal{A}\ \mathrm{dt}$ — in two realizations of the same framework. The redshift formula and the phase formula are not separate postulates; they are two evaluations of the same quantity in different state spaces. Formally: the standard derivation is a composition of two maps (GR result) ∘ (QM result). The present derivation is a single map: (distinguishability geometry) → (observable phase difference). The formula $\Delta\varphi = 2\pi\nu(gh/c^2)T$ is not new. The unification of its derivation is new.

Dephasing in an atomic ensemble. For an ensemble of atoms with a Gaussian distribution of heights σ_h (e.g. a Sr lattice clock with finite cloud extent), each atom accumulates a different $\varDelta T_G^{GR}$. The ensemble-averaged Ramsey contrast is:

$$C(T) = C_0 \cdot \langle e^{i\varDelta\varphi} \rangle_{\sigma_h} = C_0 \cdot exp\,[-\frac{1}{2}(2\pi\nu \cdot \frac{g\sigma\_h}{c^2} \cdot T)^2] \tag{B4}$$

where: $C(T)$ — Ramsey fringe contrast as a function of interrogation time T, C_0 — initial contrast at $T = 0$, σ_h — r.m.s. height spread of the atomic ensemble; for a 1 mm Sr cloud $\sigma_h \sim 0.3$ mm, $\langle e^{i\varDelta\varphi} \rangle$ - ensemble average over the Gaussian height distribution; yields the Gaussian envelope. Physical interpretation is the atoms at different heights accumulate different GR geometric times $\varDelta T_G^{GR}$; the spread in $\varDelta T_G^{GR}$ causes dephasing of the Ramsey fringes

Equivalently, taking the logarithm:

$$-\ln \frac{C(T)}{C_0} = \frac{1}{2}(2\pi\nu \cdot \frac{g\sigma_h}{c^2} \cdot T)^2 \tag{B5}$$

This is the gravitational distinguishability dephasing formula. In the present framework it is derived from the spread in accumulated GR geometric time across the ensemble, without additional assumptions. In the standard approach it requires separately invoking the gravitational redshift from GR and the Gaussian averaging from statistical QM.

B.2. Numerical Estimates and Experimental Status

For a strontium optical lattice clock with standard parameters [6]:

Table 5 Parameters of Sr optical clocks [6].

Parameter	Value	Source
ν_{Sr} (clock frequency)	4.3×10^{14} Hz	The $^1S_0 \rightarrow {}^3P_0$ optical clock transition in Sr
$\Delta_n = hv$	2.84×10^{-19} J	Clock-transition energy gap
σ_h (1 mm cloud)	0.3×10^{-3} m	Typical Sr lattice cloud size
σ_h (1 cm cloud)	3×10^{-3} m	Extended Sr cloud (JILA 2024 setup)
σ_φ at T = 1 s (1 mm)	$2\pi v\cdot g\sigma\_h/c^2 \approx 1.4\times10^{-8}$ rad	Phase spread per second
σ_φ at T = 1000 s (1 mm)	$\approx 1.4\times10^{-5}$ rad	Integrated over 1000 s
T* (1/*e* contrast time, 1 mm)	$T^* = c^2/(2\pi v\cdot g\ \sigma_h) \approx 70\ 000$ s	Time for C to fall to 1/ *e*
T* (1 cm cloud)	≈7 000 s	Accessible with longer interrogation
Current Sr clock coherence time	∼100–1000 s	Limited by laser linewidth, not GR dephasing

The dephasing time $T^* \sim 7000$–70 000 s is far beyond current interrogation times (~100 s), so gravitational distinguishability dephasing has not been observed. However, two experimental directions make it accessible:

Possible experiment 1: Large clouds at long interrogation times. A Sr cloud with $\sigma_h = 1$ cm interrogated for $T = 1000$ s gives $\sigma_\varphi \approx 1.4\times10^{-4}$ rad, corresponding to a fractional contrast loss of $\sim10^{-8}$. This is at the edge of current sensitivity ($\delta\varphi \sim 10^{-4}$ rad in state-of-the-art setups). The

BECCAL apparatus on the ISS (launch planned 2026) works with large BEC clouds at extended interrogation times in microgravity; a vertical gradient could be applied deliberately.

Possible experiment 2: Sr cloud with controlled height spread. The JILA group (Ye lab) has demonstrated gravitational redshift measurement in a 1 mm Sr cloud [16]. The next step — not yet performed — is to measure the Ramsey contrast (not just the frequency) as a function of cloud size σ_h and interrogation time T, and verify the quadratic dephasing law $-\ln(C/C_0) \propto \sigma_h^2\, T^2$. This is a direct experimental test of the gravitational distinguishability dephasing formula.

B.3 Why This Is Not Just Standard QM + GR

A sceptical reader may object that the gravitational Ramsey dephasing formula derived above follows directly from standard QM + GR without the distinguishability framework. This is correct: the formula is the same. The conceptual difference is in the derivation (see Table 6)

Table 6 Conceptual differences

Standard derivation (QM + GR)	Present derivation (distinguishability)
Step 1: Invoke GR → $\Delta\nu/\nu = gh/c^2$ (Einstein redshift postulate)	Single postulate: ΔT_G^{GR}= $(gh/c^2)\cdot T$ from geodesic deviation in $\mathscr{F}$_GR
Step 2: Invoke QM → $\Delta\varphi = 2\pi\nu\cdot\Delta\nu\cdot T$ (Ramsey phase postulate)	Same object: ΔT_G^{QM}= T in $\mathscr{F}_{QM}$; phase = $2\pi\nu\Delta T_G^{GR}$
Step 3: Combine externally → $\Delta\varphi = 2\pi\nu(gh/c^2)T$	Equality $\Delta T_G^{GR} = \Delta T_G^{QM}$ at measurement event → same formula
Two separate theories, combined ad hoc	One geometric principle, two realizations
Cannot predict what happens when $\mathcal{A}_{QM} \neq \mathcal{A}_{GR}$	Predicts new observable when $\mathcal{A}_{QM} \neq \mathcal{A}_{GR}$

B.4. New Prediction: Contrast Shift from Accessibility Mismatch

The distinguishability framework does make a genuinely new prediction when the QM accessibility ratio $\mathcal{A}_{QM}$ is externally modified while the GR accessibility ratio $\mathcal{A}_{GR}$ is unchanged. This occurs when the Ramsey pulse intensity (Rabi frequency Ω_R) is varied: a stronger drive increases v_{FS} and hence $\mathcal{A}_{QM}$, but does not change the gravitational geodesic deviation.

In the standard theory, changing Ω_R does not affect the Ramsey phase accumulated during the free-evolution period — the phase is determined entirely by the frequency offset, independent of the pulse intensity. In the distinguishability framework, the geometric time during the Ramsey pulses contributes to the total T_G, and this contribution depends on Ω_R. The correction to the phase is:

$$\Delta\varphi_{total} = \Delta\varphi_0 \cdot (1 + \beta \cdot \mathcal{A}_{QM}^2) = 2\pi\nu \cdot \frac{gh}{c^2} \cdot T \cdot (1 + \beta \cdot \frac{\hbar\Omega_R^2}{\Delta_n^2}) \tag{B7}$$

where: $\Delta\varphi_0 = 2\pi\nu(gh/c^2)T$ — standard Ramsey phase from gravitational redshift, β - dimensionless coupling constant from the distinguishability action S_{dist}; to be determined from the variational calculation , $\hbar\Omega_R/\Delta_n$ — ratio of Rabi energy to clock-transition energy gap; equals $\mathcal{A}_{QM}$ for the two-level system during the pulse . Measurability for $\Omega_R = 2\pi\times10$ Hz and $\Delta_n = h\nu_{Sr}$: $\hbar\Omega_R/\Delta_n \sim 10^{-14}$; correction $\beta\times10^{-28}$ is unmeasurable at current sensitivity. Path to measurability requires system with $\mathcal{A}_{QM} \sim 10^{-2}$ to 10^{-3}; e.g. driven quantum processor in orbit (Section 8 future directions)

The immediate value of this prediction is not experimental but structural: it shows that the distinguishability framework has a distinct empirical content beyond standard QM + GR. When $\mathcal{A}_{QM}$ can be tuned independently of $\mathcal{A}_{GR}$, the two frameworks give different numbers. The standard theory predicts no Ω_R-dependence of the Ramsey phase during free evolution; the distinguishability framework predicts a correction of order $\mathcal{A}_{QM}^2$. This is a falsifiable distinction.

The gravitational Ramsey dephasing formula derived in Section 7a warrants a careful discussion of its conceptual status. The formula itself $\Delta\phi = 2\pi\nu\left(\frac{gh}{c^2}\right)T$and its ensemble average $C(T) \approx C_0 \exp\left[-\frac{1}{2}(2\pi\nu g\sigma_h T/c^2)^2\right]$ is not new. It follows from combining the Einstein gravitational redshift with the standard Ramsey phase relation, and has been implicitly assumed in all gravitational clock comparison experiments since Pound and Rebka [17].

What is new is the structure of the derivation. The standard approach requires two independent inputs from two separate theories: the redshift formula $\Delta\nu/\nu = gh/c^2$is imported from general relativity [12], and the phase-accumulation relation $\phi = 2\pi\nu t$ is imported from quantum mechanics. These are then combined externally — the GR result is substituted into the QM formula by hand. The two theories are not unified; they are glued.

The distinguishability geometry framework produces the same formula from a single primitive structure. The geodesic deviation of two worldlines separated by height $h$in a gravitational field generates accumulated GR geometric time $\Delta T_G^{GR} = (gh/c^2) \cdot T$, computed entirely within the GR realization of distinguishability geometry with accessibility scale $\Lambda_{GR} = c/\hbar$. Independently, a resonantly driven two-level atom accumulates QM geometric time $\Delta T_G^{QM} = T$, computed within the QM realization with $\Lambda_{QM} = \Delta_n/\hbar$. The Ramsey phase difference is the product of the clock frequency and the difference in accumulated geometric times between the two interferometer arms. No separate postulate about redshift or quantum phase is required — both enter as evaluations of the same geometric object $T_G = \int \mathcal{A}\, dt$in different state spaces.

This structural unification has a precise formal content. The standard derivation is a composition of two independent mappings: GR → $\Delta\nu/\nu$ followed by QM → $\Delta\phi$. The distinguishability derivation is a single mapping: distinguishability geometry → observable phase difference. The number of independent postulates is thereby reduced from two to one.

It is important to state clearly what this does and does not imply. The unification of the derivation does not, by itself, constitute a new physical prediction — the predicted phase $\Delta\phi$is the same in both approaches. The new physical content appears only when the two accessibility ratios $\mathcal{A}_{QM}$and $\mathcal{A}_{GR}$are independently varied. In the standard framework, the Ramsey phase accumulated during free evolution is independent of the drive intensity (Rabi frequency Ω_R) — the phase is set entirely by the frequency offset. In the distinguishability framework, the geometric time accumulated during the Ramsey pulses carries a contribution proportional to $\mathcal{A}_{QM}^2 = (\hbar\Omega_R/\Delta_n)^2$, generating a correction $\Delta\phi_{total} = \Delta\phi_0 \cdot (1 + \beta\mathcal{A}_{QM}^2)$to the total accumulated phase.

For strontium optical clocks operating at $\Omega_R = 2\pi \times 1$Hz with clock transition gap $\Delta_n = h\nu_{Sr} = 2.84 \times 10^{-19}$J, the accessibility ratio during the Ramsey pulse is $\mathcal{A}_{QM} \approx 1.2 \times 10^{-15}$, and the correction is of order $\beta \times 10^{-30}$— far below any foreseeable experimental sensitivity. The reason is that optical atomic clocks are precision instruments precisely because they operate deep in the adiabatic regime $\mathcal{A}_{QM} \ll 1$, which suppresses all distinguishability corrections.

Appendix C. YIG Magnonic Resonator Realization

This appendix outlines a possible realization of distinguishability geometry in nonlinear magnonic resonators based on yttrium iron garnet (YIG). The central idea is that nonlinear magnon systems provide direct access to strongly non-adiabatic evolution, making them a natural platform for investigating geometric-time accumulation. Consider a YIG resonator driven by an external microwave pumping field. The system evolves through a sequence of magnonic states characterized by mode amplitudes, phases, and occupation numbers. Within the distinguishability framework, neighboring magnonic configurations are represented as neighboring points in the distinguishability manifold. The corresponding distinguishability distance is $d\ell^2 = G_{AB}\, dq^A dq^B$. The geometric velocity is $v_G = d\ell/dt$. The accumulated geometric time is $T_G = \int \mathcal{A}\, dt$, where $\mathcal{A} = v_G / \Lambda$. The central prediction is that two different pumping protocols producing identical accumulated geometric times should exhibit equivalent distinguishability signatures, even when the microscopic mode populations evolve differently. Experimentally, distinguishability may be quantified through measurable observables such as:

- magnon occupation number;
- phase coherence;
- nonlinear frequency shift;
- mode-conversion efficiency;
- parametric excitation probability.

A direct test of the theory consists of constructing T_G for different pumping protocols and determining whether the measured observables collapse onto a common curve when plotted as a function of T_G rather than laboratory time. Observation of such a collapse would indicate that geometric time provides a more fundamental description of the evolution than the external laboratory clock. Failure of the collapse would falsify the proposed distinguishability description for this platform. Because YIG resonators naturally operate in strongly nonlinear and non-adiabatic regimes, they provide an especially sensitive test of the distinguishability framework.

Appendix D. Bose–Einstein Condensate Realization

This appendix discusses a possible realization of distinguishability geometry in Bose–Einstein condensates (BECs). BEC systems are particularly attractive because they provide coherent many-body quantum states whose evolution can be monitored with high precision. Within the distinguishability framework, a condensate trajectory corresponds to a path in the manifold of physically distinguishable many-body states. The local distinguishability distance is $d\ell^2 = G_{AB}$ $dq^A dq^B$. The corresponding geometric velocity is $v_G = d\ell/dt$. The accumulated geometric time is $T_G = \int \mathcal{A}\ dt$, where $\mathcal{A} = v_G / \Lambda$. Possible observables include:

- condensate fraction;
- phase coherence;
- interference visibility;
- population imbalance;
- collective-mode amplitudes.

The theory predicts that condensate evolutions generated under different experimental conditions should exhibit common behavior whenever the accumulated geometric times are equal.

Specifically, if two experiments satisfy $T_G^{(1)} = T_G^{(2)}$, then observables associated with distinguishability accumulation should collapse onto universal curves despite differing microscopic parameters. This prediction differs from conventional approaches, which generally compare systems through laboratory time, interaction strength, or particle number. The distinguishability framework instead predicts that accumulated geometric time is the relevant variable governing evolution.

A particularly promising test involves matter-wave interferometry. Interference visibility provides a direct measure of distinguishability between condensate states and therefore offers a natural experimental probe of geometric-time accumulation. Observation of universal behavior after geometric-time normalization would provide evidence that distinguishability geometry captures a common structure underlying many-body quantum evolution. Failure of such universality would place strong constraints on the validity of the framework.